CPPC 21/L

DIGITAL SIGNAL PROCESSING

ACTIVITY No.2

RGB Image Processing

Submitted by:

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BS CpE 3B

Submitted to:

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Subject Professor

I. Objectives

- To examine the RGB values of a selected image using MATLAB.
- To understand how red, green, and blue values form the color of an image.
- To extract the RGB channels from an image.
- To transfer the RGB values into Microsoft Excel.
- To use conditional formatting in Excel to visualize color intensity.
- To compare the processed RGB output with the original image.

II. Materials

- MATLAB
- Laptop
- Sample Image
- Microsoft Excel
- Microsoft PowerPoint

III. Procedures

1. Open MATLAB

- Launch the MATLAB application.
- Click New Script to create a new script file.
- Upload or place the sample image in the working folder.
- In the script, type the command for reading the image:

```
stitch = imread("stitch.png");
```

- Go to the Command Window to continue checking the image values.

2. Load the Image

- In the Command Window, type the command below:

```
img = imread(stitch.png ')
```

- This command loads the image into MATLAB.
- The image data will be stored in the variable named img.

3. Extract the RGB Channels

- After loading the image, separate the red, green, and blue channels.
- Type the following commands in the Command Window:
    - R = img(:,:,1);
    - G = img(:,:,2);
    - B = img(:,:,3);
- The variable R contains the red color values.
- The variable G contains the green color values.
- The variable B contains the blue color values.

4. Copy the RGB Values

- Highlight the RGB values shown in the Command Window.
- Right-click the selected values.
- Choose Copy.
- Make sure the copied values are complete before transferring them to Excel.

5. Paste the Values in Excel

- Open Microsoft Excel.
- Click an empty worksheet.
- Paste the copied RGB values into the sheet.
- Arrange the pasted data properly so it can be formatted and analyzed.

6. Separate the Data into Columns

- Highlight the pasted values in Excel.
- Go to the Data tab.
- Click Text to Columns.
- Choose the delimiter option.
- Select space as the delimiter.
- Click Finish.
- The values should now be separated into different columns.

7. Apply Conditional Formatting

- Highlight the RGB values in the Excel worksheet.
- Go to the Home tab.
- Click Conditional Formatting.
- Select Color Scales.
- Click More Rules if the color scale needs to be adjusted.
- This will help show the intensity of the RGB values visually.

8. Adjust the Color Scale

- Set the minimum and maximum values based on the RGB value range.
- Use the appropriate color scale depending on the selected RGB channel.
- Apply the formatting to show the variation of color intensity.
- Repeat the process for the red, green, and blue values if needed.

9. Compare the Result with the Original Image

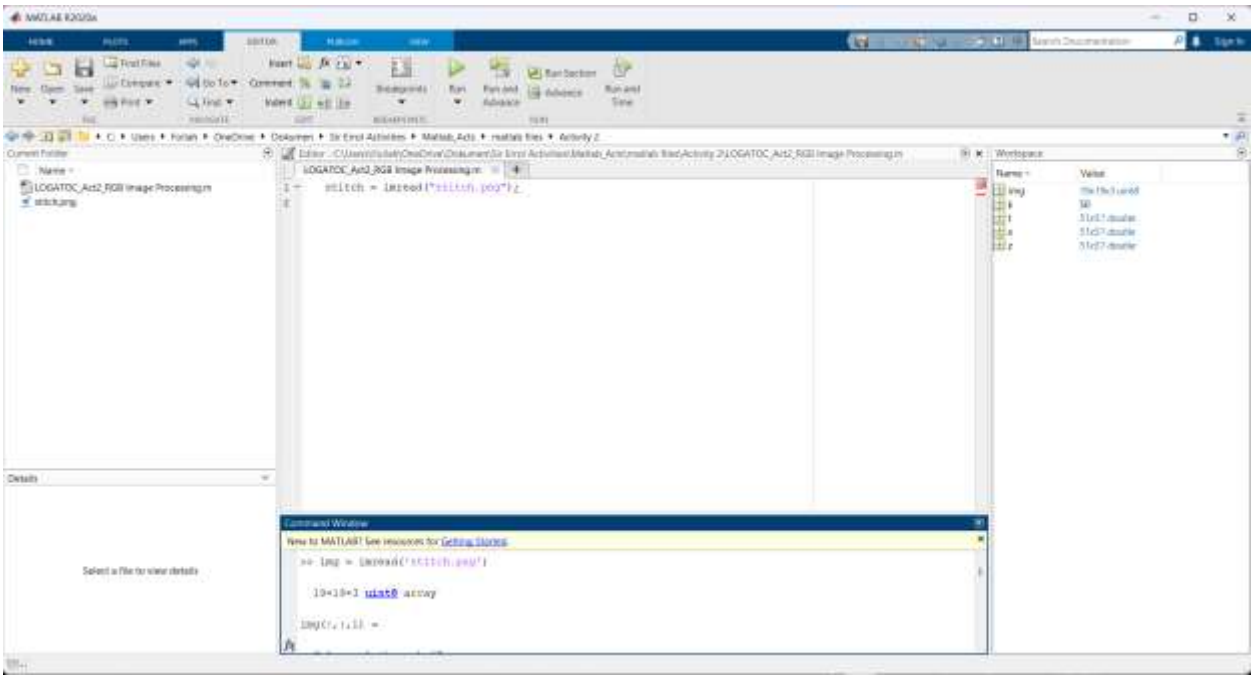
- Observe how the RGB values appear after applying conditional formatting.

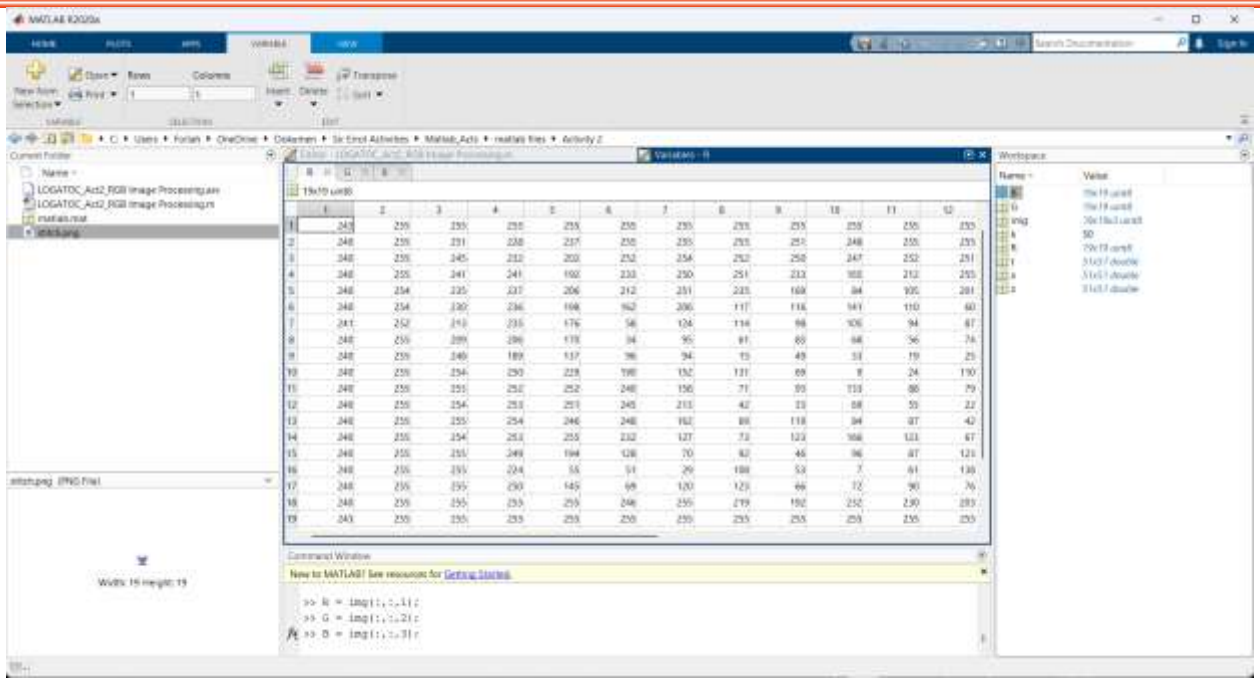
- Take screenshots of the RGB value outputs in Excel.
- Paste the screenshots into Microsoft Word.
- Adjust the transparency of the RGB outputs.
- Compare the processed RGB visuals with the original image.
- Check how the red, green, and blue channels contribute to the final image appearance.

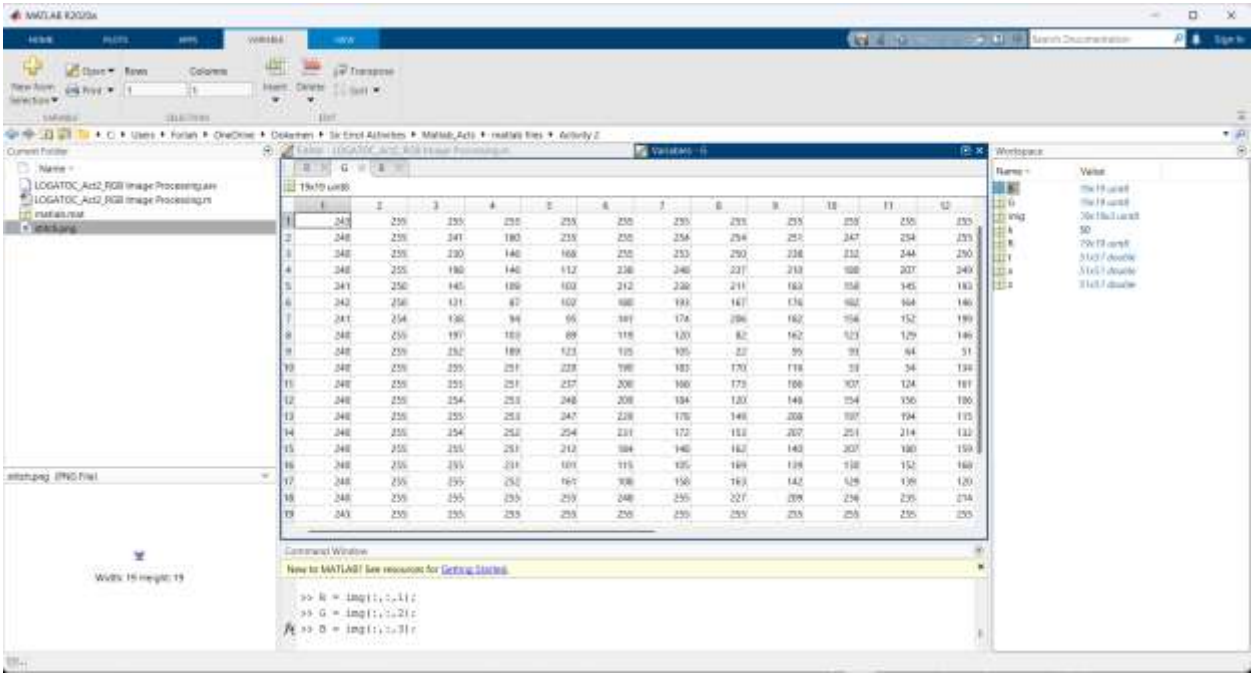
IV. Conclusion

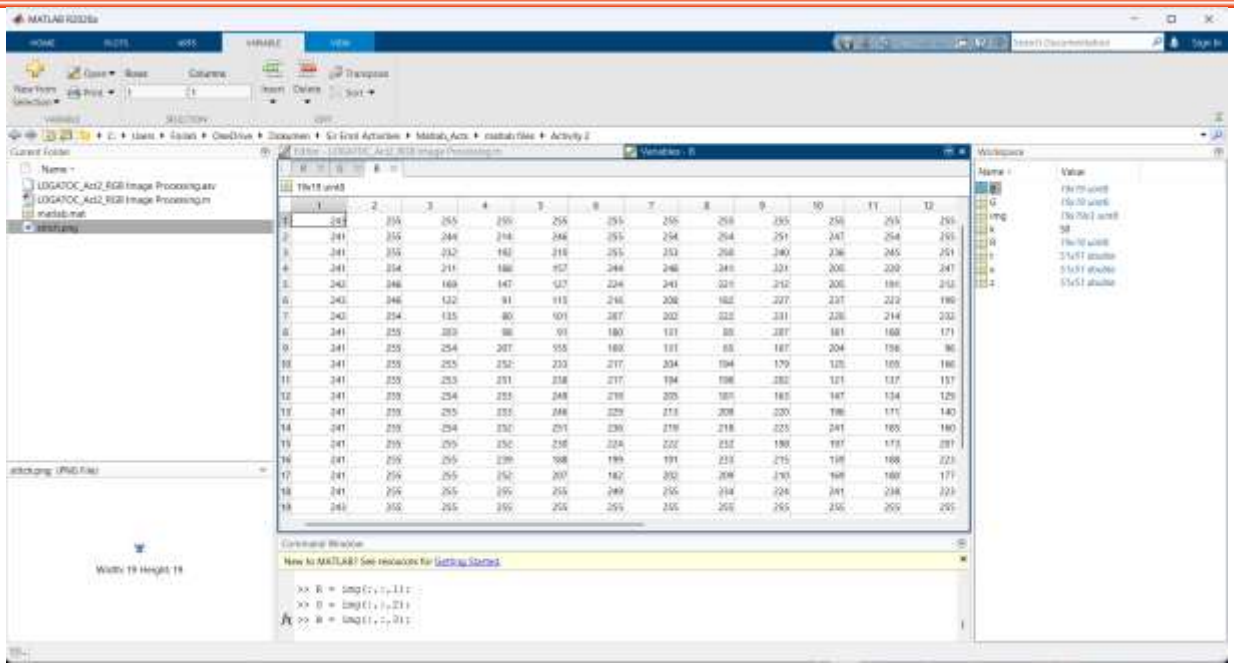
The activity showed how MATLAB can be used to extract RGB pixel values from an image and how these values can be analyzed using Excel. By separating the red, green, and blue channels, the image was represented through numerical color values. After transferring the data to Excel and applying conditional formatting, the differences in color intensity became easier to understand visually. Overall, this activity helped demonstrate how RGB components work together to form an image and how digital images can be processed through MATLAB and spreadsheet visualization.

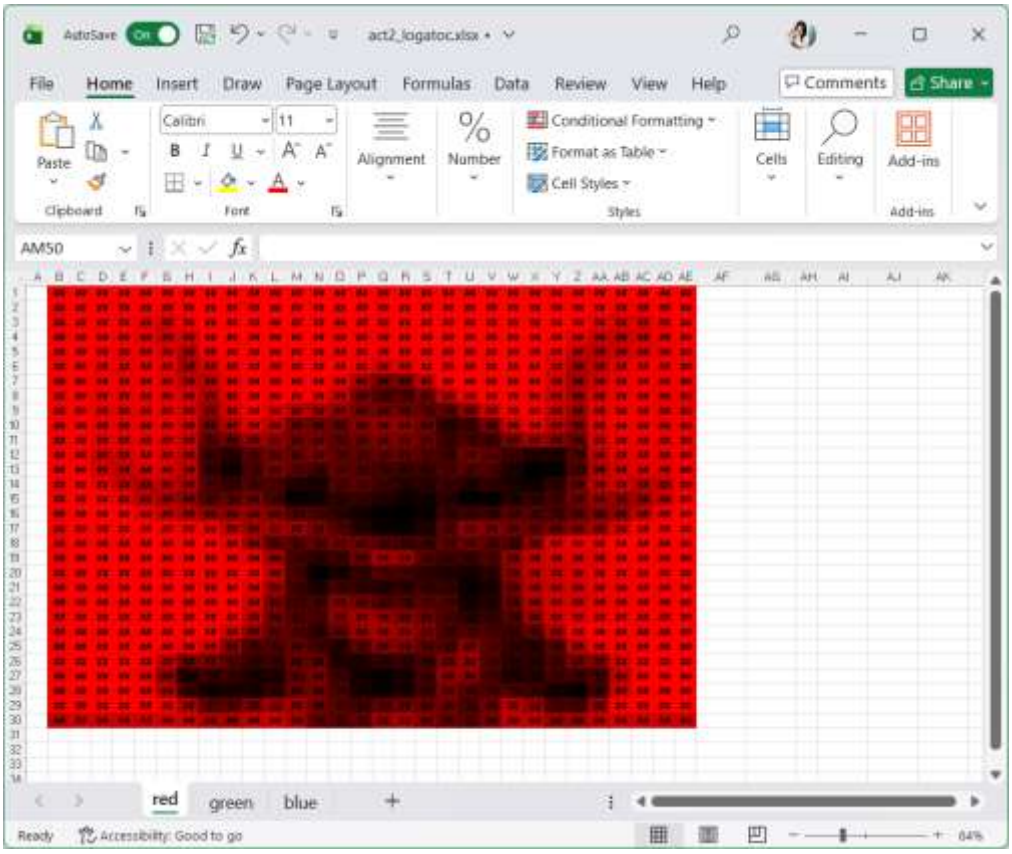
V. Documentation




MATLAB R2020a
Variables - R
19x10 uint8
Command Window
>> R = img(:,:,1);
>> G = img(:,:,2);
>> B = img(:,:,3);


MATLAB R2020a
Variables - G
19x10 uint8
Command Window
>> R = img(:,:,1);
>> G = img(:,:,2);
>> B = img(:,:,3);

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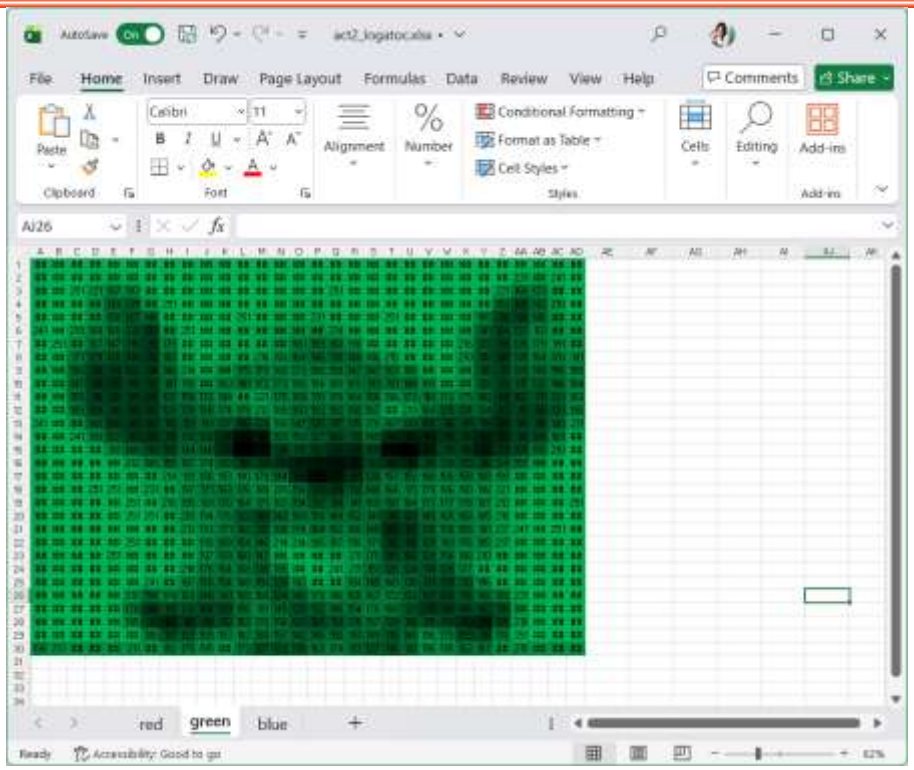
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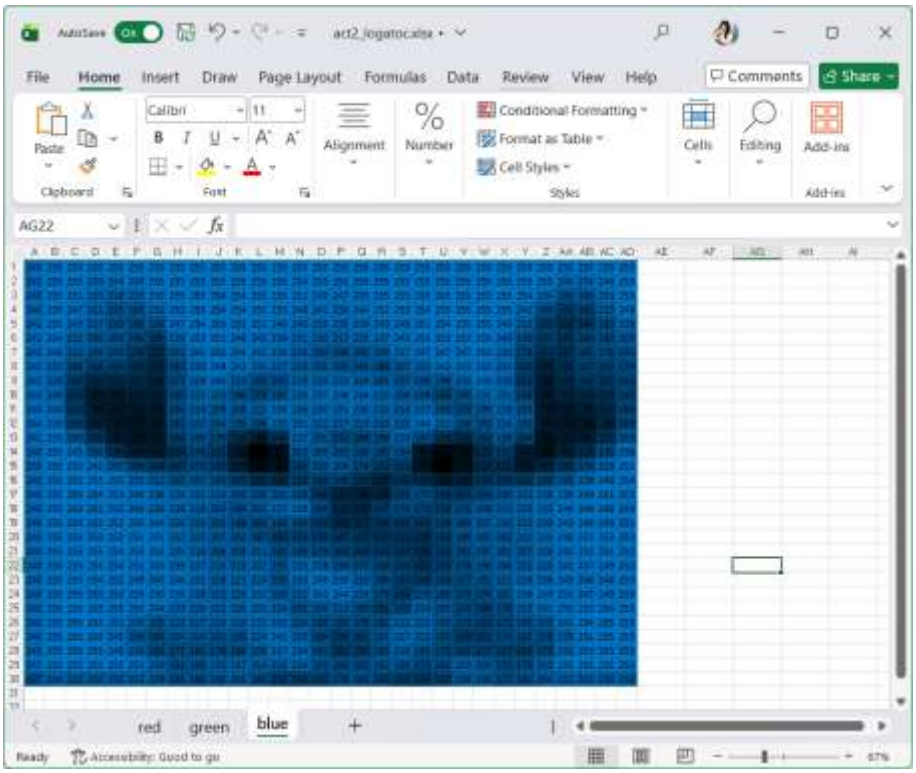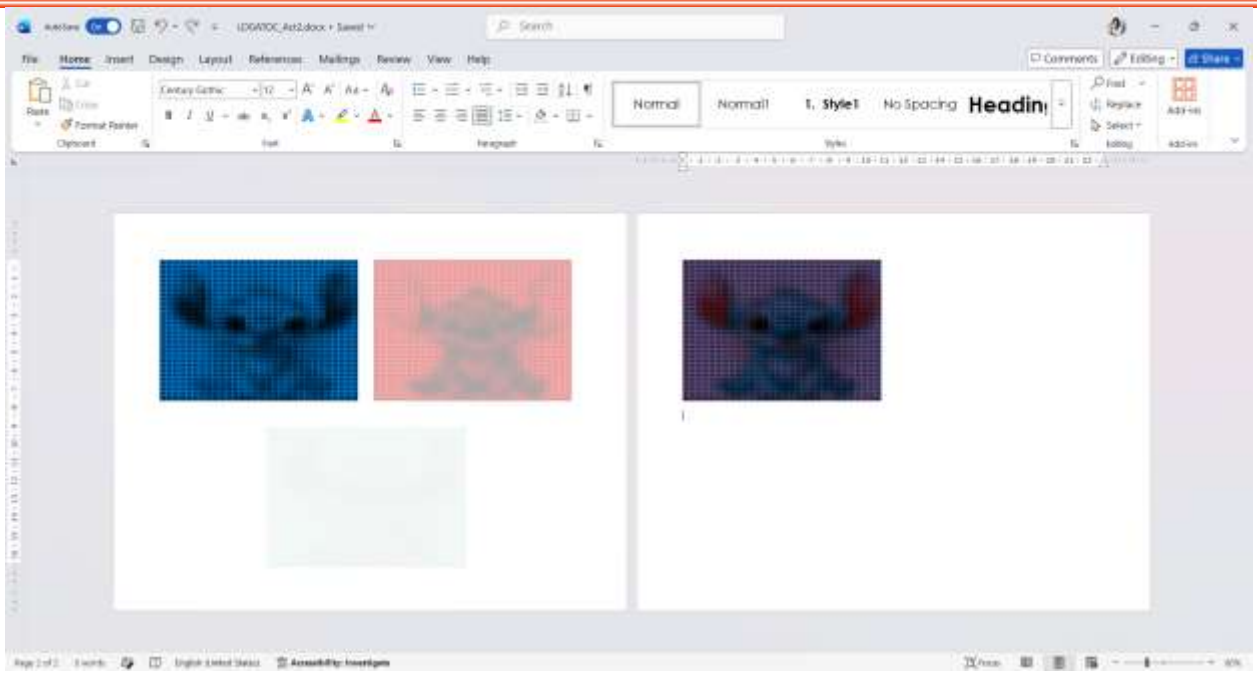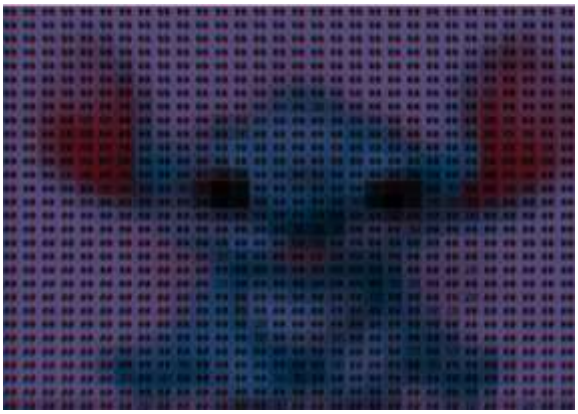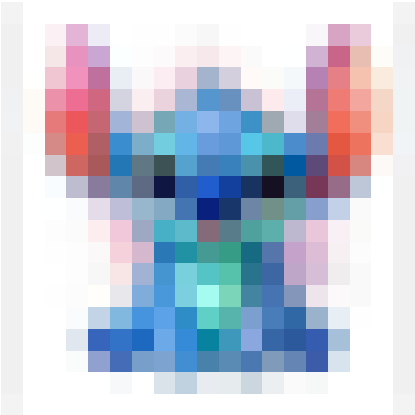
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